

OpenEMR Messaging Flow Guide

A complete walkthrough of the **Messages**, **Reminders**, and **Recalls** features — who sends, who receives, what gets emailed to the patient, and where the data is stored.

Feature	Who can use it	Who receives it	Does the patient see it?
Messages Top menu → Messages	STAFF only	STAFF mailbox	NO — internal only
Reminders Patient chart → Reminders tab	STAFF only	STAFF mailbox	NO — patient is just the SUBJECT
Recalls Patient chart → Recalls tab	STAFF creates entry	CRON mails/SMSes patient	YES — via email/SMS if configured
Patient Portal Messages Top menu → Messages → To: patient	STAFF or PATIENT	The other side	YES — stored in portal + email notification (after our patch)

Common confusion: The screens labelled "Messages" and "Reminders" inside a patient chart look like they message the patient, but they do not — they message *other staff members* about that patient. To actually contact the patient, use **Patient Portal Messages** or **Recalls**.

1. Messages (Staff-to-Staff)

Internal task/message inbox between OpenEMR users (providers, billers, front desk, etc.). Optionally pinned to a patient chart for context. The patient is not notified in any way.

Flow diagram

Staff user A
writes message



DB table
pnotes



Staff user B
sees inbox

No email. No SMS. No portal notification. Strictly internal.

What's in the screen

- **To:** dropdown lists *users* from the **users** table.
- **Patient:** optional link to a chart (just for context, not a recipient).
- **Type:** categorization (Unassigned, Lab, Bill, etc.).
- **Date:** when it was sent.
- **Status:** New / Read / Done.

Code & storage

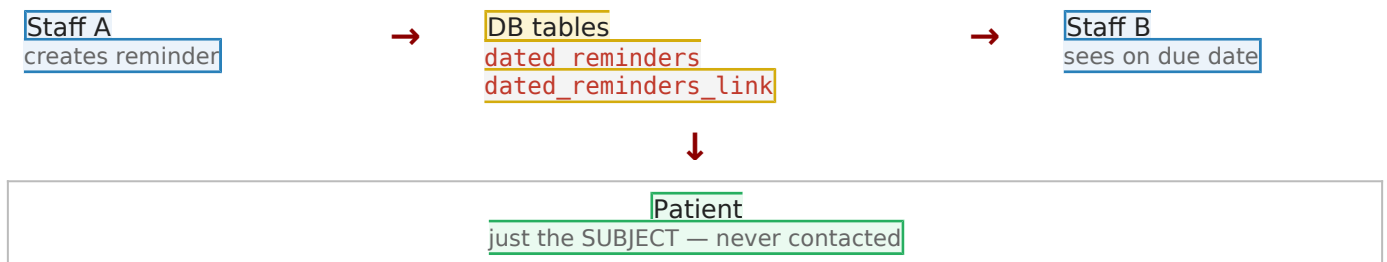
Item	Location
UI handler	<code>interface/main/messages/messages.php</code>
Helper functions	<code>library/pnotes.inc.php</code>

Storage table	pnotes
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2. Reminders (Dated Reminders — Staff Internal)

A scheduled internal reminder system. Staff member A creates a dated reminder about Patient X and assigns it to Staff member B (or themselves). On the due date, it shows up in B's reminder list. **The patient is never contacted.**

Flow diagram



Schema (truncated)

```

dated_reminders
  dr_id          -- PK
  dr_from_ID     -- users.id (creator)
  dr_message_text -- the note
  dr_message_due_date -- when it pops up
  pid            -- patient subject (NOT recipient)
  message_priority -- 1=Low, 2=Med, 3=High

dated_reminders_link
  dr_id -- FK
  to_id -- users.id (recipient staff member)
  
```

Code

Item	Location
Add-reminder form	interface/main/dated_reminders/dated_reminders_add.php
List view	interface/main/dated_reminders/dated_reminders.php
Log view	interface/main/dated_reminders/dated_reminders_log.php

3. Recalls (Patient Outreach via MedEx)

A recall is a follow-up reminder for the **patient** — "come in for your annual physical," "your lab work is due," etc. Unlike Messages and Reminders, recalls **do** reach the patient, via email, SMS, or a printed letter. This is run by the **MedEx** module on a cron schedule.

Flow diagram



What MedEx needs to function

1. **Email gateway** — SMTP configured under Admin → Globals → Notifications (**set this up first**)
2. **SMS gateway** — Twilio or Clickatell credentials in the Fax/SMS module (paid services, no free option)
3. **Background service active** — Admin → Background Services → enable the MedEx job
4. **Patient has contact info on file** — Demographics → Email / Phone
5. **Patient opted in** — `patient_data.hipaa_allowemail`, `hipaa_allowsms`

Schema (truncated)

```
medex_recalls
  r_ID          -- PK
  r_pid         -- the patient receiving the recall
  r_eventDate   -- when the recall is due
  r_provider    -- the provider attached
  r_reason      -- short text reason
  r_facility    -- which clinic

medex_outgoing -- queue of actual messages to send
notification_log -- record of what went out
```

4. Patient Portal Messages (Two-way Secure Messaging)

The proper way to actually message a patient from inside OpenEMR. Stored as a "secure message" in the portal mailbox; an email notification tells the patient there's something new to read. **The message body itself is not emailed** — the patient must log in to read it (HIPAA-safe).

Flow diagram (after our patch)



What we patched on 2026-05-23

In `portal/lib/portal_mail.inc.php`, the `sendMail()` function now calls a new helper `queuePortalMailPatientNotification()` when the recipient copy of a message is stored. The helper looks up the patient's email via `patient_access_onsite` → `patient_data` and drops a notification into the `email_queue` table. The actual sending is done by the **EmailService** background job. Failures never block the in-portal save (wrapped in try/catch).

Required setup

Setting (Admin → Globals → Notifications)	Value
Email Transport Method	SMTP
SMTP Server Hostname	smtp.gmail.com
SMTP Server Port Number	587
SMTP User for Authentication	your Gmail address

SMTP Password for Authentication	16-char Gmail App Password
SMTP Security Protocol	TLS
Patient Reminder Sender Email	same Gmail address

Patient prerequisites

1. Patient exists in Demographics with an **Email** populated
2. Patient has portal credentials — Demographics → Login Details → "Allow Patient Portal"

5. End-to-end Test — Verify a Portal Message Reaches a Patient

1. Log in to OpenEMR as **admin**
2. Open the patient chart → Demographics → confirm Email is set
3. Demographics → Login Details → ensure portal account exists
4. Top menu → **Messages** → + **Add New**
5. Set **To:** the patient (not a staff user)
6. Type subject + body → **Send**
7. In phpMyAdmin run:

```
SELECT id, recipient, subject, sent, error_message, datetime_queued
FROM email_queue ORDER BY id DESC LIMIT 5;
```

You should see a new row.

8. Admin → Background Services → trigger **EmailService** (or wait for its interval)
9. Check the patient's Gmail inbox for "New secure message in your patient portal"

If something fails

Symptom	Likely cause & fix
No row in email_queue	Patient has no email on file, no patient_access_onsite row, or patient_reminder_sender_email global is blank.
email_queue.error_message populated	Gmail rejected. Re-check App Password and 2FA on the Google account.
Row stays sent=0 forever	EmailService background job is disabled or interval too long. Admin → Background Services → activate.
PHP error log shows <i>Portal mail notification failed</i>	Read c:\xampp\php\logs\php_error_log for full stack — usually SMTP misconfig.

6. Quick Reference Card

Question	Answer
"I clicked Send in the Messages screen — will the patient get it?"	NO . Internal staff message only.
"I clicked Send in the Reminders screen — will the patient get it?"	NO . Internal staff reminder only.

"I created a Recall for the patient — will they get notified?"	YES , on the next MedEx cron cycle, by email/SMS/letter depending on patient prefs.
"I sent a Patient Portal Message — will they get an email about it?"	YES , after our patch and SMTP configuration. They get a "new message" alert; the body is read in the portal.
"Where is SMTP configured?"	Admin → Globals → Notifications tab. Stored in globals table. Password is encrypted.
"Where do I see what went out?"	email_queue (portal/recall notifications), notification_log (MedEx history).